JU HOSPITAL, JAIPUR

MEDICAL GAS PIPELINE SYSTEM (MGPS)

Technical Specification and Bill of Quantities, Revision 05

Project	JU Hospital, Jaipur
Prepared by	Acenzo Planning & Architecture LLP
Issued by	JECRC Foundation (private entity, project management)
Scope	Supply, installation, testing and commissioning of complete MGPS: O_2, N_2O, CO_2, Medical Air 4 bar, Medical Air 7 bar, Vacuum, AGSS
Standards	ISO 7396-1:2016, EN 13348, BS EN 1254-1, IS 3316, European Pharmacopoeia (medical air quality)
Document Status	Issued for Tender, Rev 05
Quantities	Reconciled with vendor following query and response process. Final quantities to be verified at site against GFC drawings before procurement.
Date	July 2026

Certification basis: All terminal units, manifold control panels, area alarm and area valve units, AGSS units, and the master alarm panel shall hold a valid safety certificate issued by a competent authority, or CE marking issued by a notified body registered with the European Commission, with the 4-digit notified body identification number permanently marked on the unit and traceable on the EC NANDO database. Acenzo reserves the right to independently verify any certificate submitted, directly with the issuing notified body where required. Any certificate that cannot be independently verified shall be rejected without further discussion.

Pre-qualification: The manufacturer shall demonstrate a minimum of five years' experience manufacturing medical gas pipeline equipment, and at least three installations of comparable scale (300 or more beds) that have been in continuous operation for a minimum of two years, with contactable client references to be furnished at pre-award stage.

Inspection and traceability: Critical active components (manifold control panels, terminal units, compressors, vacuum pumps) shall be subject to a factory acceptance test witnessed by the Architect's representative or an appointed third-party inspection agency prior to dispatch. Mill test certificates for copper and brass raw material shall be submitted, batch traceable. One physical sample of each terminal unit type and area alarm unit shall be submitted for inspection and written approval before bulk manufacture.

Vendor neutrality: This specification is performance-based and does not name or favour any manufacturer or brand. Equipment of any origin, domestic or imported, is acceptable provided the certification, pre-qualification, and inspection requirements above are met in full.

1 Project Overview and System Summary

JU Hospital, Jaipur is a private tertiary-care teaching hospital, approximately 10,00,000 sq ft across B+G+9 floors in three blocks (A, B, C) with 1000+ beds, managed by JECRC Foundation as Project Management Consultant. The MGPS covers all clinical areas from Basement to upper floors including ICUs, OTs, emergency, maternity, imaging, day care, wards, and procedure rooms. The system shall be designed, supplied, installed, tested, and commissioned in full compliance with ISO 7396-1:2016 and the standards listed below.

Standard	Application	Notes
ISO 7396-1:2016	Primary standard governing MGPS, plant, distribution, terminal units, alarms, and commissioning. Full compliance mandatory.	
EN 13348	European standard for copper tubes for medical gases and vacuum. All copper pipe to be manufactured, tested, and certified to this standard.	
BS EN 1254-1	Capillary solder fittings for copper pipe. All pipe fittings to be end-feed capillary type certified to this standard, performance equivalent acceptable from any manufacturer.	
IS 3316	Indian standard for medical gas systems, referenced for any locally applicable requirements.	
European Pharmacopoeia	Medical air quality standard. Purified air supplied must meet European Pharmacopoeia specification for medical use.	
Medical Device Regulation (CE)	Mandatory for all critical components per the certification clause above: manifold panels, terminal units, area alarms, AGSS, master alarm panel.	

2 Terminal Unit Schedule, Room-wise (Reconciled)

Note: *Quantities below reflect the room-wise schedule as confirmed correct by the vendor in response to the query list dated June 2026, with the following corrections applied: AGSS removed from LINAC, PET-CT, Gamma Camera, MRI and CT rooms (A1, A2, A11, A12, B11-B14) per vendor confirmation that exposure duration does not warrant scavenging; D4 Dental X-Ray corrected to O2 and Vacuum only; D5 Dental Chair set to zero centralized requirement as the unit is self-contained; E21 IVF OT has N2O removed as this room is dedicated to IVF use. All figures remain subject to verification at site against GFC drawings before procurement.*

Ref	Shaft	Room / Area	Beds	O_2	Air 4	N_2O	Vac	Air 7	AGSS	CO_2
A	**BASEMENT LEVEL**									
A1	SHAFT-1	LINAC RM.-1	1	2			2			
A2	SHAFT-1	LINAC RM.-2	1	2			2			
A3	SHAFT-1	BRACHYTHERAPY RM.	1	2			2		1	
A4	SHAFT-1	BRACHYTHERAPY HOLDING RM.	2	2			2			
A5	SHAFT-1	LINAC RM. HOLDING RM.	2	2			2			
A6	SHAFT-1	PET-CT POST DOSE RM.-1,2,3	3	3			3			
A7	SHAFT-1	PET-CT POST DOSE DISPENSING	1							
A8	SHAFT-1	DOSE ADMIN PET-CT	1							
A9	SHAFT-1	DOSE ADMIN GAMMA CAMERA	1							
A10	SHAFT-1	GC POST DOSE SPACE RM.- 1,2	2	2			2			
A11	SHAFT-1	GAMMA CAMERA	1	2			2			
A12	SHAFT-1	PET -CT	1	1	1		1			
				18	**1**		**18**		**1**	
B	**LOWER GROUND FLOOR**									
B1	SHAFT-1	EMERGENCY WARD	24	48			24			
B2	SHAFT-1	POST OP.	3	6	3		3			
B3	SHAFT-1	ISOLATION ROOM - 01 & 02	2	4	2		4			
B4	SHAFT-1	MINOR OT	1	2		1	2		1	
B5	SHAFT-1	EMERGENCY O.T- 01	1	4	3	2	5	2	1	2
B6	SHAFT-1	EMERGENCY O.T- 02	1	4	3	2	5	2	1	2
B7	SHAFT-1	TRIAGE	3	6			6			
B8	SHAFT-1	TREATMENT ROOM	1	1			1			
B9	SHAFT-1	PLASTER ROOM	1	1			1			
B10	SHAFT-1	RESUSCITATION	9	18	18		18			
B11	SHAFT-2	MRI RM.	1	2	1	1	2			
B12	SHAFT-2	CT RM.	1	2	1	1	2			
B13	SHAFT-2	MRI RM.	1	2	1	1	2			
B14	SHAFT-2	CT RM.	1	2	1	1	2			
B15	SHAFT-2	X-RAY-1,2,3,4.	4	4			4			
B16	SHAFT-2	MAMMOGRAPHY RM.	1							
B17	SHAFT-2	HOLDING AREA (RADIOLOGY)	2	2			2			

Ref	Shaft	Room / Area	Beds	O_2	Air 4	N_2O	Vac	Air 7	AGSS	CO_2
B17	SHAFT-2	ULTRASOUND- 1,2,3,4.	4	4			4			
B18	SHAFT-2	MATERNITY HDU/ICU	12	24	12		24			
B19	SHAFT-2	NICU-1	12	24	24		24			
B20	SHAFT-2	NICU-2	11	22	22		24			
B21	SHAFT-2	NURSERY	11	22	22		22			
B22	SHAFT-2	POST-PARTUM	12	12			12			
B23	SHAFT-2	LABOUR ROOM	3	6	3	3	6			
B24	SHAFT-2	SEPTIC RECOVERY	2	2			2			
B25	SHAFT-2	SEPTIC LABOUR ROOM	1	1	1	1	1			
B26	SHAFT-2	ECLAMPSIA	2	4	2		4			
B27	SHAFT-2	POST-OP/RECOVERY	4	4	2		4			
B28	SHAFT-2	FIRST STAGE ROOM/ PRE OP	9	9			9			
B29	SHAFT-3	TSSU	1							
B30	SHAFT-3	C-SEC OT	1	4	3	2	5		1	2
B31	SHAFT-3	C-SEC OT	1	4	3	2	5		1	2
				250	**127**	**17**	**229**	**4**	**5**	**8**
C	**UPPER GROUND FLOOR**									
C1	SHAFT-1	TMT RM.	1	1			1			
C2	SHAFT-1	ECG RM.	1	1			1			
C3	SHAFT-1	ECHO RM.	1	1			1			
C4	SHAFT-1	PFT	1	1			1			
C5	SHAFT-1	INJECTION ROOM (M)	1	1			1			
C6	SHAFT-1	INJECTION ROOM (F)	1	1			1			
C7	SHAFT-1	MINOR OT- ORTHO	1	2			2		1	
C8	SHAFT-	DRESSING ROOM (M & F)	2	3			2			
C9	SHAFT-3	PROCEDURE ROOM (OPHTHAL)	1	1			2			
C10	SHAFT-3	LASER ROOM (DERMA)	1	1			1			
C11	SHAFT-3	MINOR OT (OPTHAL)	1	2			2		1	
C12	SHAFT-3	DRESSING ROOM SURGERY (M&F)	2	2			2			
C13	SHAFT-3	MINOR OT (ENT)	1	2			2		1	
C14	SHAFT-3	MINOR OT (SURGERY)	1	2			2		1	
C15	SHAFT-3	PROCEDURE ROOM (DERMA)	1	1			1			
C16	SHAFT-3	VITALS+INJECTION	2	2			2			
C17	SHAFT-3	ENG. LAB	1	1	1		1			
				25	**1**		**24**		**4**	
D	**FIRST FLOOR**									
D1	SHAFT-1	MINOR OT-OBG	1	2			2		1	
D2	SHAFT-1	PFT	2	2	2		2			
D3	SHAFT-1	VITALS+INJECTION	1	1						
D4	SHAFT-1	PROCEDURE ROOM- OBG	1	1			1			

Ref	Shaft	Room / Area	Beds	O_2	Air 4	N_2O	Vac	Air 7	AGSS	CO_2
D5	SHAFT-1	ULTRASOUND- 1	1	1			1			
D6	SHAFT-2	TREATMENT ROOM	2	2			2			
D7	SHAFT-2	BRONCHOSCOPY	1	2	1	1	2		1	
D8	SHAFT-2	ULTRASOUND- EHC	1	1			1			
D9	SHAFT-2	ECHO- EHC	1	1		1				
D10	SHAFT-2	TMT- EHC	1	1						
D11	SHAFT-2	ECT ROOM- PSYCHIATRY	1	2		1	2		1	
D12	SHAFT-2	EEG ROOM- PSYCHIATRY	1	1			1			
D13	SHAFT-2	TREATMENT ROOM- PSYCHIATRY	1	1			1			
D14	SHAFT-3	GENERAL WARD (OBSTETRICS & GYNAECOLOGY)	30	30			30			
D15	SHAFT-3	TREATMENT RM.	1	1			1			
D16	SHAFT-3	GENERAL WARD (OBSTETRICS & GYNAECOLOGY)	60	60			60			
D17	SHAFT-3	TREATMENT RM.	2	2			2			
				110	**3**	**3**	**108**		**3**	
E	**SECOND FLOOR**									
E1	SHAFT-1	CATH LAB- 01	1	2	2	1	2		1	
E2	SHAFT-1	CTVS O.T	1	4	3	2	5	2	1	2
E3	SHAFT-1	CATH WASH CUM SCRUB	1		2			2		
E4	SHAFT-1	CATH LAB- 02	1	2	2	1	2		1	
E5	SHAFT-1	CATH HOLDING	7	7			7			
E6	SHAFT-1	CCU & CTVS- ICU	31	62	31		62			
E7	SHAFT-2	DAYCARE (ONCOLOGY)(27 BEDS)	27	27			27			
E8	SHAFT-2	DIALYSIS (13+7 BEDS) + ISO.	20	20			20			
E9	SHAFT-2	PROCEDURE ROOM	1	2			2			
E10	SHAFT-2	ENDO ROOM-01	1	2	1	1	2		1	
E11	SHAFT-2	ENDO ROOM-02	1	2	1	1	2		1	
E12	SHAFT-2	ENDO ERCP ROOM	1	2	1	1	1		1	
E13	SHAFT-2	ENDO WASH	1		2			2		
E14	SHAFT-2	ENDO INSTRUMENT WASH	1	1			1			
E15	SHAFT-2	ENDO PRE PROCEDURE	2	2			2			
E16	SHAFT-2	ENDO POST PROCEDURE	2	2			2			
E17	SHAFT-2	ENDO USG	1	1			1			
E18	SHAFT-2	MINOR PROCEDURE ROOM	1	2			2		1	
E19	SHAFT-2	USG- LITHOTRIPSY	1	1			1			
E20	SHAFT-2	URO- HOLDING AREA	1	2			2			
E21	SHAFT-2	ECHO	1	1			1			
E22	SHAFT-2	ECG	1	1			1			
E23	SHAFT-2	BRONCHOSCOPY	1	2	1	1	2		1	
E24	SHAFT-2	PFT	1				1			

Ref	Shaft	Room / Area	Beds	O_2	Air 4	N_2O	Vac	Air 7	AGSS	CO_2
E25	SHAFT-3	GENERAL WARD (MALE) + TREATMENT		31			31			
E26	SHAFT-3	GENERAL WARD (FEMALE) - SURGERY)	31	31			31			
E27	SHAFT-3	GENERAL WARD (MALE) -SURGERY) +	31	31			31			
				243	**46**	**8**	**242**	**6**	**8**	**2**
F	**THIRD FLOOR**									
F1	SHAFT-1	OT-1	1	4	3	2	5	2	1	2
F2	SHAFT-1	OT-2	1	4	3	2	5	2	1	2
F3	SHAFT-1	STERILE STORE/ TSSU	1					1		
F4	SHAFT-1	OT-3	1	4	3	2	5	2	1	2
F5	SHAFT-1	OT-4	1	4	3	2	5	2	1	2
F6	SHAFT-1	OT-5	1	4	3	2	5	2	1	2
F7	SHAFT-1	OT-6	1	4	3	2	5	2	1	2
F8	SHAFT-1	OT-7	1	4	3	2	5	2	1	2
F9	SHAFT-1	STERILE STORE/ TSSU	1					1		
F10	SHAFT-1	OT-8	1	4	3	2	5	2	1	2
F11	SHAFT-1	OT-9	1	4	3	2	5	2	1	2
F12	SHAFT-1	ROBOTIC ARM O.T	1	4	3	2	5	2	1	2
F13	SHAFT-1	POST-OP	20	20	20		20			
F14	SHAFT-2	PRE-OP	20	20	20		20			
F15	SHAFT-2	ICU 01	10	20	10		20			
F16	SHAFT-2	ICU 02	10	20	10		20			
F17	SHAFT-2	ICU 03	10	20	10		20			
F18	SHAFT-3	GENERAL WARD (PSYCHIATRY) + TREATMENT R	21	21			21			
F19	SHAFT-3	GENERAL WARD (SURGERY) + TREATMENT RM.	31	31			31			
F20	SHAFT-3	GENERAL WARD (SURGERY) + TREATMENT RM.	31	31			31			
				223	**100**	**20**	**233**	**22**	**10**	**20**
G	**FOURTH FLOOR**									
G1	SHAFT-1	CSSD			4			4		
G2	SHAFT-1	BLOOD CENTER		6			6			
G3	SHAFT-2	ICU 04	18	36	18		36			
G4	SHAFT-2	BURN ICU	6	12	6		12			
G5	SHAFT-3	GENERAL WARD (MALE) + TREATMENT RM.	31	31			31			
G6	SHAFT-3	GENERAL WARD (FEMALE) + TREATMENT RM.	31	31			31			
G7	SHAFT-3	GENERAL WARD (FEMALE) + TREATMENT RM.	31	31			31			
				147	**28**		**147**	**4**		
H	**FIFTH FLOOR**									

Ref	Shaft	Room / Area	Beds	O_2	Air 4	N_2O	Vac	Air 7	AGSS	CO_2
H1	SHAFT-1	TWIN SHARING IPD ROOMS	24	24			24			
H2	SHAFT-1	SINGLE IPD ROOM	18	36			18			
H3	SHAFT-2	SUPERSPECIALITY WARD-1 + TREATMENT RM.	31	31			31			
H4	SHAFT-2	SUPERSPECIALITY WARD-2 + TREATMENT RM.	31	31			31			
H5	SHAFT-2	SUPERSPECIALITY WARD-3 + TREATMENT RM.	31	31			31			
H6	SHAFT-2	SUPERSPECIALITY WARD-4 + TREATMENT RM.	31	31			31			
H7	SHAFT-3	GENERAL WARD (DERMA) + TREATMENT RM.	21	21			21			
H8	SHAFT-3	GENERAL WARD (FEMALE) + TREATMENT RM.	31	31			31			
H9	SHAFT-3	GENERAL WARD (MALE) + TREATMENT RM.	31	31			31			
				267			**249**			
I	**SIXTH FLOOR**									
I1	SHAFT-1	TWIN SHARING IPD ROOMS	24	24			24			
I2	SHAFT-1	SINGLE IPD ROOM	18	36			18			
I3	SHAFT-2	SUPERSPECIALITY WARD-5 + TREATMENT RM.	31	31			31			
I4	SHAFT-2	SUPERSPECIALITY WARD-6 + TREATMENT RM.	31	31			31			
I5	SHAFT-2	SUPERSPECIALITY WARD-7 + TREATMENT RM.	31	31			31			
I6	SHAFT-2	SUPERSPECIALITY WARD-8 + TREATMENT RM.	31	31			31			
I7	SHAFT-3	GENERAL WARD (PEDIATRICS) + TREATMENT RM.	21	21			21			
I8	SHAFT-3	GENERAL WARD (PEDIATRICS) + TREATMENT RM.	31	31			31			
I9	SHAFT-3	GENERAL WARD (PEDIATRICS) + TREATMENT RM.	31	31			31			
				267			**249**			
J	**SEVENTH FLOOR**									
J1	SHAFT-1	SINGLE IPD ROOM	16	32			16			
J2	SHAFT-1	SINGLE IPD ROOM	6	12			6			
J3	SHAFT-3	GENERAL WARD (OPHTHALMOLOGY) + TREATMENT RM.	21	21			21			
J4	SHAFT-3	GENERAL WARD (ORTHOPEDICS) + TREATMENT RM.	31	31			31			
J5	SHAFT-3	GENERAL WARD (ORTHOPEDICS) + TREATMENT RM.	31	31			31			
				127			**105**			
K	**EIGHTH FLOOR**									
K1	SHAFT-3	GENERAL WARD (OTO-RHINO) + TREATMENT RM.	21	21			21			
Grand Totals (reconciled, post-vendor confirmation)				**1698**	**306**	**48**	**1625**	**36**	**30**	**30**

3 Plant and Equipment BOQ

Certification, pre-qualification, and inspection requirements for all items below are as stated in the notice on the cover page. No brand or make is specified or implied anywhere in this section.

Item	Description & Performance Specification	Unit	Qty	Remarks
1 Oxygen (O2) Supply System				
1.01	Fully automatic manifold control panel for oxygen supply. Non-electric cylinder switching, automatic changeover between primary and reserve bank and different sources of Oxygen supply (manifold and LMO) minimum flow rate 2,000 LPM, BMS compatible, compliant with ISO 7396-1.	Nos	2	*Primary and reserve banks*
1.02	Cylinder manifold, 2x24 cylinder configuration, complete with tail pipes, non-return valves, pressure relief valve, and cylinder fixing chains.	Set	2	
1.03	Emergency reserve control panel regulator for oxygen system.	Nos	2	
1.04	Cylinder manifold, 1x24 cylinder configuration, complete with tail pipes, non-return valves, pressure relief valve, and cylinder fixing chains.	Set	2	
2 Nitrous Oxide (N2O) Supply System				
2.01	Fully automatic manifold control panel for N2O supply. Non-electric cylinder switching, automatic changeover, minimum flow rate 600 LPM, BMS compatible, compliant with ISO 7396-1.	Nos	1	
2.02	Cylinder manifold, 2x16 cylinder configuration, complete with tail pipes, non-return valves, pressure relief valve, and cylinder fixing chains.	Set	1	
2.03	Emergency reserve control panel regulator for N2O system.	Nos	1	
2.04	Cylinder manifold, 1x16 cylinder configuration, complete with tail pipes, non-return valves, pressure relief valve, and cylinder fixing chains.	Set	1	
3 Carbon Dioxide (CO2) Supply System				
3.01	Fully automatic manifold control panel for CO2 supply. Non-electric cylinder switching, automatic changeover, minimum flow rate 600 LPM, BMS compatible, compliant with ISO 7396-1.	Nos	1	
3.02	Cylinder manifold, 2x4 cylinder configuration, complete with tail pipes, non-return valves, pressure relief valve, and cylinder fixing chains.	Set	1	
3.03	Emergency reserve control panel regulator for CO2 system.	Nos	1	
3.04	Cylinder manifold, 1x4 cylinder configuration, complete with tail pipes, non-return valves, pressure relief valve, and cylinder fixing chains.	Set	1	
4 Medical Air Compressor Plant, Pentaplex				
4.01	Oil-Free type medical air compressors, pentaplex configuration (5 nos., one duty, remainder standby per ISO 7396-1 sizing). Each unit air-cooled, minimum FAD 120 CFM, operating pressure 13 kg/cm2, motor rating 22 kW, suction filter and no-load start regulator. Integrated automatic PLC-based control panel managing duty/standby rotation, fault alarms, and BMS interface. Compact sound-absorbing enclosure.	Set	1	*5 nos. compressors as one pentaplex set with shared control panel*
4.02	Air receiver vessel, 3,000 litres, with pressure gauge, safety valve, drain valve, and all standard accessories.	Nos	2	
4.03	Medical air purification unit, five-stage filtration and adsorption drying, achieving medical air quality per European Pharmacopoeia. Minimum 2 nos. trains for duty/standby operation.	Set	2	*Duty/standby configuration*
4.04	Pressure reducing station with two outputs, 4 bar for medical equipment and 7 bar for surgical equipment, with pressure gauges, safety valves, and isolation valves.	Nos	2	

Item	Description & Performance Specification	Unit	Qty	Remarks
5 Medical Vacuum Plant, Pentaplex				
5.01	Medical vacuum pumps, pentaplex configuration (5 nos., rotary vane type, directly driven). Minimum capacity 2,000 LPM FAD at -450 mmHg, motor rating 7.5 kW per pump, IP 54 protection class. Fully automatic with PLC-based control panel, duty/standby rotation, fault alarms, and BMS interface.	Set	1	*5 nos. vacuum pumps as one pentaplex set*
5.02	Vacuum receiver vessel, 3,000 litres, vertical, with vacuum gauge, isolation valve, and all standard accessories.	Set	2	
5.03	Vacuum bacteria filters (3 nos.) and secretion traps (3 nos.), complete set with all connecting hardware and mounting provisions.	Set	1	
6 Electrical Control Panel				
6.01	Integrated electrical control panel for 5 air compressors and 5 vacuum pumps. Includes MCBs, MCCBs, contactors, auto-manual switches, PLC controller, BMS interface, and status indication for all plant items.	Set	1	
7 Area Alarm And Area Valve Units, Flush Type				
7.01	Area alarm and shut-off unit, single flush-mounted unit combining area alarm panel and area isolation valve, 1 gas (O2) and 1 vacuum. Compliant with ISO 7396-1.	Pcs	15	*Indicative, confirm against final room count*
7.02	Area alarm and shut-off unit, 2 gases (O2, Air 4) and 1 vacuum.	Pcs	30	*Indicative, confirm against final room count*
7.03	Area alarm and shut-off unit, 3 gases (O2, Air 4, N_2O) and 1 vacuum.	Pcs	16	*Indicative, confirm against final room count*
7.04	Area alarm and shut-off unit, 4 gases (O2, Air 4, N2O, CO2) and 1 vacuum.	Pcs	3	*Indicative, confirm against final room count*
7.05	Area alarm and shut-off unit, 5 gases (O2, Air 4, N2O, Air 7, CO2) and 1 vacuum.	Pcs	23	*Indicative, confirm against final room count*
8 Terminal Units				
8.01	Terminal unit, Oxygen (O2). Compliant with ISO 9170-1, colour-coded and gas-specific indexed to prevent cross-connection.	Pcs	1058	*As per reconciled TU schedule*
8.02	Terminal unit, Nitrous Oxide (N2O). Compliant with ISO 9170-1.	Pcs	38	*As per reconciled TU schedule*
8.03	Terminal unit, Medical Air 4 bar. Compliant with ISO 9170-1.	Pcs	375	*As per reconciled TU schedule*
8.04	Terminal unit, Medical Air 7 bar (surgical air). Compliant with ISO 9170-1.	Pcs	26	*As per reconciled TU schedule, includes OT runs and CSSD/wash area drying points*
8.05	Terminal unit, Vacuum. Compliant with ISO 9170-1.	Pcs	1058	*As per reconciled TU schedule*
8.06	Terminal unit, Carbon Dioxide (CO2). Compliant with ISO 9170-1.	Pcs	23	*As per reconciled TU schedule*
9 Master Plant Alarm Panel				
9.01	Master plant alarm panel monitoring all gas supply systems and vacuum plant. Audible and visual alarms, BMS interface, compliant with ISO 7396-1.	Set	1	
10 AGSS, Anaesthetic Gas Scavenging System				
10.01	Venturi-type anaesthetic gas scavenging system terminal unit, wall-mounting type with probe. Compliant with ISO 7396-2. AGSS uses a separate probe design from other terminal units.	Pcs	28	*As per reconciled TU schedule, OTs and procedure rooms only*

Item	Description & Performance Specification	Unit	Qty	Remarks
11 Isolation Valves, Ball Type, Dezincification-Resistant Brass				
11.01	Ball valve, 15mm bore, full bore, dezincification-resistant brass body, gas-service rated, fugitive emission tested.	Pcs	26	
11.02	Ball valve, 22mm bore.	Pcs	29	
11.03	Ball valve, 28mm bore.	Pcs	17	
11.04	Ball valve, 35mm bore.	Pcs	14	
11.05	Ball valve, 42mm bore.	Pcs	5	
11.06	Ball valve, 54mm bore.	Pcs	5	
11.07	Ball valve, 76mm bore.	Pcs	5	
11.08	Ball valve, 108mm bore.	Pcs	6	
12 Probes				
12.01	Gas-specific indexed probes for all terminal unit types except AGSS, which has a separate probe design.	Nos	2578	*As per reconciled TU schedule*

4 Copper Piping BOQ

Note: *All copper pipe to be manufactured and certified to EN 13348. Pipe to be degreased for oxygen service with hydrocarbon content on surface less than 20 mg per square metre. Quantities below are indicative from design development stage and to be verified at site before procurement. A 10 percent contingency over the stated lengths is recommended at tender stage.*

Item	Pipe Size (OD x Wall Thickness)	Approx. Qty	Unit	Remarks
1.01	Copper pipe, 12mm OD x 0.7mm wall	5,914	Mtr	
1.02	Copper pipe, 15mm OD x 0.9mm wall	10,263	Mtr	
1.03	Copper pipe, 22mm OD x 0.9mm wall	9,430	Mtr	
1.04	Copper pipe, 28mm OD x 0.9mm wall	2,136	Mtr	
1.05	Copper pipe, 35mm OD x 1.2mm wall	1,083	Mtr	
1.06	Copper pipe, 42mm OD x 1.5mm wall	530	Mtr	
1.07	Copper pipe, 54mm OD x 1.5mm wall	521	Mtr	
1.08	Copper pipe, 76mm OD x 1.5mm wall	476	Mtr	
1.09	Copper pipe, 108mm OD x 1.5mm wall	654	Mtr	
Total Indicative Pipe length		**31,007**	**Mtr**	

5 Installation Specification

The specification below governs all installation works under this contract. Deviations require written approval from the Project Architect before execution. Where a performance requirement is stated, equipment or hardware of any manufacture meeting that requirement is acceptable; no make is specified or implied.

Parameter	Requirement
Copper pipe standard	EN 13348. All pipe permanently and durably marked at regular intervals with the harmonised standard number EN 13348, nominal dimensions (OD x wall thickness), temper designation per EN 1173, manufacturer's identification, date of production (year and month), and confirmation of degreasing for oxygen. Certified to EN 13348:2008.
Pipe material temper	Tail pipe: R220. Pipe OD 12mm to 54mm: R250 (half-hard). Pipe OD 76mm and above: R290 (hard). Material: Cu + Ag 99.90%, P 0.015% to 0.040%.
Pipe degreasing	Pipes degreased to ensure hydrocarbon content on surface below 20 mg/m2. Fittings degreased to ensure hydrocarbon content below 100 mg/m2. Documentation required at pre-award stage.
Pipe fittings	End-feed capillary fittings to BS EN 1254-1:1998, dezincification-resistant brass, solder maximum cadmium content 0.025% by weight. Any manufacturer meeting this standard and the certification clause is acceptable.
Pipe supports	Galvanised steel pipe saddles at maximum 1,200mm centres for pipe up to 28mm OD, at maximum 1,500mm centres for pipe 35mm and above, with adequate documented load rating for the pipe weight at full liquid or test pressure. Supports to isolate pipe from direct contact with structural elements using approved isolation liners.
Fasteners	Expansion anchors or screw-in anchors suited to the substrate (concrete, masonry, or framing as applicable), with a documented load rating exceeding three times the maximum pipe weight at full liquid or test pressure.
Brazing process	All joints brazed using oxygen-acetylene flame only, no flux, no LPG or other fuel gas. Copper to copper: silver-copper-phosphorous brazing alloy, minimum 5% silver content. Copper to brass: silver brazing alloy, minimum 43% silver content.
Nitrogen purge during brazing	All pipe continuously purged with dry nitrogen (minimum purity 99.99%) throughout the entire brazing process, from initiation of heat to complete cooling of the joint to ambient temperature. No purge interruption permitted.
Pipe identification	All pipework colour-coded and labelled per ISO 32 and ISO 7396-1 requirements at all branches, valves, penetrations, and at maximum 3-metre intervals in ceiling voids and plant rooms.

6 Testing and Commissioning

Note: *No gas shall be introduced into the pipeline system before the cross-connection test has been completed, documented, and accepted by the Project Incharge. This is a non-negotiable hold point.*

Test / Activity	Requirement
Initial pressure test, all systems	All pipework pressure-tested with dry nitrogen or oil-free dry air to 1.5x working pressure. Hold for minimum 24 hours with calibrated pressure gauge. Zero pressure drop required to pass. Test records documented with gauge calibration certificates.
Leak test, all joints	Full system leak test with helium or approved leak detection fluid at working pressure. Each joint individually checked. Any leak rectified and full test repeated.
Cross-connection test	Performed per ISO 7396-1 Annex F before terminal units are installed. Mandatory prior to energising the system with any medical gas.
Particulate contamination test	Pipeline flushed with oil-free dry nitrogen. Particulate count at terminal units to comply with ISO 7396-1.
Flow and pressure verification	Measured flow rate and pressure at each terminal unit under simultaneous use conditions, complying with ISO 7396-1 minimum requirements for the respective gas and area.
Area alarm verification	All area alarm panels tested for correct alarm activation at set pressures, correct gas identification, audible and visual signal, and BMS relay output.
AGSS functional test	All AGSS terminal units tested for correct scavenging function per ISO 7396-2.
Handover documentation	Full commissioning dossier submitted at handover: as-installed drawings (pipe routes, valve positions, terminal unit schedule by room), all test reports with dates and calibration references, equipment manuals, certificates for all critical items per the certification clause, and maintenance schedule.

End of Document - JU Hospital, Jaipur | MGPS Tender and BOQ | Rev 05 | Prepared by Acenzo Planning & Architecture LLP | Issued by JECRC Foundation | July 2026